The a_0 Law with Cold Dark Energy Counted Once: the Form the Record Closes On, Its Gates, and What Stays Open

Working note (PAPER72), version 1.1, 2026-11-10 (v1.0: DOI 10.5281/zenodo.24106867). AI-assisted research programme; not peer reviewed; not reproduced by any outside group. Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (v2.1, DOI 10.5281/zenodo.24092073, field equations and the record since) and PAPER71 (v1.1, DOI 10.5281/zenodo.24095262, the switch class). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

This note states the form on which the record closes and grades it. It does not complete the theory. The a_0 law alone fails the CMB, the growth of structure, clusters and the Bullet, each for want of the same cold, clustering component of about 5.4 times the baryons. Every way tried of keeping that component out of galaxies has a failing verdict; the last one is recorded here. The form chosen by the owner on 2026-11-09 counts the component once: it is in galaxies too, and around condensed baryons the law describes how it sits. It is real mass, not a second force. In the adopted action this is the static state's own reading: the law's halo is the khronon sector's energy density, carried by the one conserved charge that is dust in the expanding background. We regrade the form with criteria frozen before any number. On groups and clusters it is not falsified, by a gate that can fail from one side only, and the richest clusters sit at its ceiling; the Bullet passes only if the subcluster's stripped gas is counted. On hosted dwarfs it fails: it removes most of the law's shortfall in every class, but two classes stay high and four low rows cannot be reached. On KiDS lensing by galaxy type the test returns no verdict; its readings put early types between the law and the form's truncated halo and cannot place discs; on host-free lenses the form passes a gate that does not separate it from the law. The form passes rotation curves. Its CMB, lensing and growth rows hold when computed, on an amended reading of that lane's checks. With the amplitude of fluctuations pinned by large-angle polarisation, CMB lensing matches the prediction to 0.3σ and the growth rate passes. Raw KiDS weak lensing, on a gravity-only spectrum, prefers an amplitude five of its own widths lower: the absolute fit is excluded at 3σ by the frozen line and reads tension once the reference run's realisation noise is corrected, a check made after grading. This is the tension that any cold component fitted to the CMB carries; the form states neither gas ejection nor neutrino mass, and does not relieve it. With gravity coupled the static state is linearly stable around a point mass; an instability found earlier belonged to a planar background.

Version 1.1 adds the lanes that landed on 2026-11-10 and states the theory in six rules (Section 7). The collapse was computed and breaks down: a clock that starts as dust folds at the centre, and the equations give no answer beyond that moment. The static state is stable around isolated discs away from their centres; a disc in an external field has no verdict, with stability excluded there, and a disc's own centre is open. No dust term tried, keyed on the clock's offset, on its expansion or on its acceleration, joins the falling state to the standing one; the clock's leafwise flux cannot supply the halo; and the action has no cluster state at the form's own dust constant. A survey of media with more structure than one clock finds no class without a failing verdict. On the data side, most of the weak-lensing gap was the survey's superseded release and our own inputs: on KiDS-Legacy, with a neutrino floor, the covariance at the model and the reference run's draw corrected, the form's amplitude sits 1.71/1.88 of the survey's widths high with an acceptable fit, while DES Y3 stays at 2.18/2.23 on a re-scanned fit. These readings rest on post hoc amendments: the lane's amended label is GAP STAYS and its frozen labels are NO VERDICT. The two failing dwarf classes do not survive the record's own systematics budget, and a gas-history reading does not explain them. The form's data rows are computed without the joint between its two states, and several of them fail or have no verdict. The form is not complete. We list what is open, the measurement that would decide each item, and our own wrong estimates.

1 Two forms

Cold dark energy is this record's name for a gravitating field component with $w \approx 0$ at the cosmic amount; it is not a particle. Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are carried throughout and never pooled; pairs of numbers are canonical/alt unless a row says otherwise.

Form A, the law alone, is what PAPER70 states. It passes rotation curves and the Solar System. It fails the CMB (which needs a cold component of 5.41 times the baryons), growth, X-COP by $+0.4675/+0.4345$ dex, the Bullet ($Z = -4.55$, the mass sits on the gas) and Coma's ultra-diffuse galaxies ($+1.12$ dex with a static external field, $+1.41$ with a moving host).

Form B, the law with cold dark energy counted once, is the single-counting reading that lanes PD1 and PD2 graded with one fixed share per halo. As graded there it is FAIL/FAIL: it passes its population budget, rotation curves (rms 0.0563/0.0558 against a line of 0.0660) and the carried cosmology, and fails KiDS's pooled outer points (worst -0.635 dex), the groups and the Bullet's amount; X-COP passes only in a reported bracket ($+0.019/+0.010$).

On 2026-11-09 the owner chose B as the form the record closes on, with both forms scored side by side from the record. The choice is of a form, not of a pass: no label was changed by it. It reopens the cold sector retired on 2026-11-01 for this one reading. Section 3 is the regrade that followed.

2 Form B in the adopted action

The action of PAPER70 has certified field equations (lane QX17). A note written after the decision, and checked independently for this paper, reads form B out of them for a static clock.

- **The law's halo is energy density.** The khronon sector's energy density contains $\Pi = D\cdot P$, and the static weak-field 00 equation is Poisson's with source $\rho_b + D\cdot P$. Light and matter both see it.
- **One conserved charge.** The khronon equation is a conserved current whose charge is, to leading order, $D\cdot P + \mu_K^2(Q-1)/K$. With $P = 0$ it is cold dust in the expanding background; around a static source it is the law's halo plus the static dust term $-\mu_K^2\Phi/K$, which is zero at $\mu_K = 0$. Nothing is counted twice.
- **The static state is in force balance.** With $J(g) = (a_0^2/K)[u - \ln(1+u)]$, $u = g/a_0$, the law sector's stress satisfies $\partial_j \Pi_{ij} = (D\cdot P)\, g_i$. In the deep field $J = g^2/2K$, which is the pressure of an isothermal halo with $\sigma^2 = \frac{1}{2}\sqrt{GM_b a_0}$: the relation PD1 had to assume. The identity holds for any $J(g)$ with a curl-free field, so it shows that the action's static state balances itself; it is not evidence for this J.
- **Force balance is not stability; the stability was then computed.** With the metric frozen (the decoupling limit) and in a planar background, lane TPA1 found the action's law-on phase gradient-unstable, with no ghost, wherever the clock's own dust density is below a critical value; the e-folding times on its grid are 5.0 Myr to 0.62 Gyr. The note on form B read the form's static state, which has almost no dust of its own, as lying in that range. Lane SCS1 then coupled gravity around a point mass and labels the state STABLE on both footings: two independent routes find no growing mode at any wavenumber, and disturbances oscillate at 0.08 to 12 times the local free-fall rate. The clock's gradient coefficient is its own charge density, $n_d + D\cdot P$. In TPA1's planar background $D\cdot P = p'$, which is negative; around a point mass $D\cdot P = p' + 2p/r$ is the phantom density, which is positive. SCS1 reproduces TPA1's numbers in the planar background, and the growth returns when that background is put back into the coupled system. TPA1's other findings stand, and the note's sentence has been corrected.
- **What that does not settle.** The result is linear, and for point masses only. For flattened baryons the law's phantom density can be negative in places, and there the same closed form gives growth; SCS1 did not analyse discs or extended spherical distributions. Version 1.1: lane SCS2 analysed discs. Isolated discs (razor-thin exponential and Kuzmin at four masses, a thick disc, a disc with a bulge) are STABLE on both footings: $D\cdot P$ is positive in every cell on both grids and no mode grows on either route. SCS2's frequency known answer fails with its frozen fit and passes under its fourth amendment, which is post hoc and followed a graded run. With a uniform external field along the axis or in the plane the label is NO VERDICT on both footings: both routes find unstable regions, and the magnitudes are not converged or rest on one grid. Lane SCS3 took the external-field case alone and also returns NO VERDICT on both footings; it says that the sign is not in doubt and that STABLE is excluded there. It also says that SCS2's label for isolated discs does not cover the centre, and that whether the static clock is stable at a disc's own centre is open. Lane DCN1 then took the centre. Every reading it froze before its outcome was seen is NO VERDICT, each on a known-answer gate; under the last of its six post hoc amendments, two of them written after the outcome was seen, it reads ARTEFACT on both footings, with the isolated centre stable. Whatever the reading, every one of its 24 resolved solves has $D\cdot P$ positive in every cell and no growing rate. Extended spherical distributions remain unanalysed. At the dust constant the CMB needs, the static state is Jeans-unstable at the local free-fall rate (e-folding times of 17 Myr to 3.9 Gyr): that is the cosmological setting and not the form's galaxy setting. The lane's three amendments are post hoc and disclosed; one mutation arm does not bite, being a field redefinition, as the lane said before any run.

What the action does not give is stated in Section 5, items 1 to 4.

3 The regrade

Each lane froze its criteria in a commit before any graded number was computed (KO2 ran a model-only design toy before its freeze and says so), and has known answers that must pass and mutation twins that must fail. KO2, RB1, RB2, CO1, CO2, HE1, TPA1 and ONE1 were each re-run by a session other than the one that wrote it, and their labels reproduced; the re-runs are listed in `closure_2026_11_09/RERUNS_2026-11-09.md` and `fable52_push/README.md`. SCS1 (Section 2) landed last: its lead session re-ran both of its routes and recompiled its Lean file (13 theorems); the first route reproduces with only timing lines differing, and the second gives every label identical and every eigenvalue equal to better than one part in a million. The lanes relabel nothing. The lanes added in version 1.1 are in Sections 2, 5 and 7, and Section 7 says which of them were re-run.

3.1 Groups, clusters and the Bullet (RB1): NOT FALSIFIED (ONE-SIDED) on both footings

In form B a hot system holds $R = 5.41$ times the baryons that fell into it, and the component stays when gas is expelled or stripped. The retained fraction $f_{\rm ret} = R\,M_{b,\rm obs}/(M_{\rm dyn} - M_{b,\rm obs})$ can therefore not exceed 1 at a system's edge. This gate can fail from above only; the two-sided amount gate stays FAIL as PD1 graded it.

- **Groups.** The 20 hydrostatic groups read $f_{\rm ret} = 0.597$ [0.556, 0.637] at R_{500}, none above 1. Form B reads the missing 40 % as gas outside R_{500}. Nothing on disk measures it.
- **Clusters sit at the ceiling.** X-COP at its last radius reads 1.095 [0.885, 1.259], with 4 of 7 clusters above 1. At R_{500} on the forward-fitted mass the class reads 1.057 [1.046, 1.072], above the ceiling beyond its errors; this row is a reported bracket, and graded there the gate would fail. On the NFW-fit hydrostatic mass the same class reads 0.959. The verdict for the deepest wells turns on about 10 % in the hydrostatic mass.
- **Trend.** $f_{\rm ret}$ rises with depth: slope +0.160 [+0.080, +0.225] in $\log M_{500}$ from groups to X-COP.
- **Radial shape.** One collisionless component with the record's own profile, anchored at each cluster's last radius, gives missing-mass fractions 0.72 / 0.63 / 0.55 / 0.47 / 0.35 per r/R_{500} bin against the measured 0.71 / 0.66 / 0.58 / 0.46 / 0.32 (canonical); bin residuals −0.007 / +0.033 / +0.006 / −0.030 / −0.041 dex against a line of 0.10. This label rests on a post hoc amendment; under the frozen known answer the gate is NO VERDICT.
- **The Bullet.** PD1's 2.6σ deficit in the subcluster is the bookkeeping of the stripped gas. With the gas shared in proportion to the stars the amounts read −0.45 / +0.37 (main / sub) and pass; with nothing reassigned they read +0.49 / −2.58 and fail. No measurement on disk fixes the subcluster's gas before the merger.

3.2 Hosted dwarfs (RB2): B FAILS ON THE HOSTED ROWS on both footings

In form B a dwarf keeps the phantom of its own baryons inside a host, cut only by tides. There is no external field effect.

- **What it removes.** Coma's ultra-diffuse galaxies move from +1.120 to +0.326 dex (alt +0.283) and read B CONSISTENT; the interval contains zero only with the systematics budget, and on statistics alone it is [+0.105, +0.547]. The Milky Way's classical dwarfs read B CONSISTENT as a class (+0.281/+0.241), with Draco, Ursa Minor and Sextans high row by row.
- **Two classes fail high.** The Milky Way's ultra-faints sit at +0.634 [+0.433, +0.865] (alt +0.592) and M31's bright dwarfs at +0.273 [+0.085, +0.473] (alt +0.231). The statistic carries measurement scatter only; a full systematics budget was not computed, since that would be a change after the freeze.
- **Rows that cut the other way.** NGC 1052-DF2 and DF4 sit 0.74 and 0.94 dex below B and at Newton on their baryons. Tides reach them only if NGC 1052 holds 17 and about 100 times the mass B gives it inside their projected distances. NGC 147 and IC 1613 sit 0.56 and 0.54 dex below B; the law with its external field leaves them as low. Crater II is the one low row tides can reach.

3.3 Lensing by galaxy type (KO2): NO VERDICT for late and early types on both footings

All 181,477 KiDS lenses were fitted with every catalogued neighbour modelled off-centre (888,795 pool objects), in place of a veto. The statistic is the ratio of the amplitude beyond the inner window to the inner amplitude;

the law as a galaxy's own field gives 1 and no extended field gives about 0.

- **Why no verdict.** One of 13 known answers failed (a cross-shear pull of 3.11 among 15 amplitudes). Without it the frozen bracket rule gives NOT SEPARATED for both types. One mutation twin did not bite: the ratio measures the shape of the outer profile, not its level.
- **Early types lie between.** They read 0.487 ± 0.071 in the primary design and 0.739 ± 0.115 with lens and neighbour amplitudes untied: below the law's level in every design and above form B's truncated halo (0.038/0.007 in the primary design).
- **Discs cannot be placed.** They read 0.277 ± 0.183 tied and 1.071 ± 0.199 untied. Form B planted with the measured cross-shear as noise reads 1.030 ± 0.282 untied, so the data cannot tell the law from form B for discs.
- **An earlier lead does not generalise.** Lane KO1, and PAPER71 beside its gate, reported that host-free late types show no outer signal (-0.19 ± 0.39 against 1.34 ± 0.18 for early types). Both labelled it a lead and not a result: seen after the freeze and sharing the veto's edge. Plain stacks of all lenses are flat for both types (late outer-to-inner ratio 0.93 ± 0.13, early 1.09 ± 0.08), though they include the neighbours' signal. With neighbours modelled, the deficit appears only inside the host-free subset (-0.03 ± 0.28), which carries the veto's edge.

A look at the host-free lenses, committed before KO2 had a lensing number, compares form B at PD1's share with KO1's shape ratio. It is post hoc and models no neighbours. At the frozen veto radius form B reads 0.510 against a measured 0.607 ± 0.094, with the law as the lens's own field 4.2σ off on the other side. At the widest veto, where the edge effect is weakest, form B reads 0.525 against 0.811 ± 0.134, 2.1σ low, with the law 1.4σ off. The note read from it that one share does not fit late and early types together; ONE1's frozen type-difference line passes at every veto, so the graded statistic does not read that. The same note predicted blind that KO2's late-type ratio would be at most about 0.5; the tied design meets that (0.277 ± 0.183) and the untied one does not (1.071 ± 0.199), so it is not confirmed. Lane ONE1 then graded form B on those lenses with every catalogued neighbour carrying the same share: KiDS host-free PASSES/PASSES by its frozen gate at the widest veto, for PD1's share and for that row only. The shape ratio there is 0.811 against a model of 1.003 (-1.14σ; alt 0.809 against 0.847); late types read 0.084 against 0.792 (-1.46σ) and early types 0.882 against 0.999. At the frozen veto radius, which carries the edge, the model misses (0.607 against 0.924, -2.68σ). The law with no end gives 0.98 at the widest veto, also inside 2σ, so the pass does not separate form B from the law, and the late-type error there (0.42) is too large to test discs. The look's 0.510 to 0.525 was mainly a matter of calibration: with the inner calibration fitted, the model reads 0.86/0.73 at the widest veto without neighbours and 1.00/0.85 with them. Re-run by its lead session: the KiDS script's label and every graded number reproduce, with only timing lines and one status flag differing; its Lean file recompiles (10 theorems) and its twin fails.

3.4 The last way of keeping the component out (HE1): FAILS on both footings

HE1 held the galaxy's share outside a wall as an envelope in pressure balance with the law's field. The envelope exists, is stable and fills 0.2 % of space or less. It fails KiDS's late-type stacks: none of its three stated shares, at any outer radius, passes its first four gates together. Post hoc and with low power, at the widest veto the record's share fits those stacks (p 0.73/0.83), as does the law. Its fifth gate fails as frozen on two clusters near R_{500} only, and the label does not rest on it. Re-run by its lead session: both labels and every number in `he1.json` reproduce; only timing lines differ. Lane TPA1 wrote out and certified the covariant equations of the two-phase action (EQUATIONS COMPLETE; health UNHEALTHY): its switch wall and its dust phase are healthy, and its law-on phase has the instability described in Section 2, which SCS1 finds does not hold around a point mass. With PAPER71's closed switch class, no reading tried that keeps the component out of galaxies passes the record's gates.

3.5 The cosmology rows, computed (CO1): ROWS HOLD on the amended reading, with an inherited halo list

The decision listed form B's cosmology rows as carried from the cold reference. A further lane computed them. Form B's linear content passes the primary CMB on raw Planck PR4 and ACT DR6 spectra ($-65.21/-65.07$ against a line of 9; cold to baryon 5.410/5.412, fitted). That row replays a fit stored by an earlier lane, and its yardstick is the likelihood package's own test point, not a framework quantity. PR4 lensing reads $0.9617 \pm 0.0231/0.9564 \pm 0.0230$ of the predicted amplitude ($-1.66/-1.89\sigma$ against a 2σ line; shape χ^2 8.56 over 9 bandpowers), and DESI DR1 $f\sigma_8$ passes (largest pull 1.22/1.23). A cold component of the fitted amount

gives the cold-component spectra, so these three rows holding is not independent evidence for the framework. For the layout frozen in the criteria (galaxies: the Lagrangian share as the law's phantom; groups and clusters: the share inside R_{200c}, part of it kept in the reference profile) the matter power changes by 0.6/1.0 % at $k = 1$ and 5.1/8.8 % at $k = 3\,h\,\mathrm{Mpc}^{-1}$, and raw KiDS-1000 cannot see it on the 104 points the reference run supports.

- **Two readings of the label.** One mutation arm did not bite as frozen; the label rests on an amendment committed before any graded number. Read as frozen it is NO VERDICT.
- **The halo list is inherited:** a mass function, a bias and a concentration relation calibrated on simulations. The redistribution rows are externally calibrated and say so.
- **The scope decides it.** If a cluster held the whole dust of its turnaround sphere, the power at $k = 1$ would be 2.9 times higher and raw KiDS would reject it. The form's cosmology needs the cluster share tied to the baryons present (5.41 times them) and counted inside R_{200c}; RB1 grades the same bookkeeping at R_{500} or the last measured radius.
- **CO1 left the amplitude open.** The fit's σ_8 is 0.8405/0.8429, with no low-multipole data to hold it. Raw KiDS fits the record's reference run at 0.811 with $p = 0.014$ and form B at its own amplitude with $p = 0.0015/0.0009$ (a bracket with constant-factor scaling, which understates it). The forest row stays carried. CO2, below, pins the amplitude and finds that these absolute p-values carry the reference run's realisation noise.

3.6 The amplitude, pinned (CO2): AMPLITUDE EXCLUDED BY RAW KiDS AT 3 SIGMA as frozen, on both footings; not robust

Lane CO2 refitted the same spectra with the Planck 2018 low-multipole polarisation table added. The optical depth moves from 0.0697/0.0729 to 0.0532 ± 0.0066 on both footings (a width along the degeneracy only, which understates the profile width) and σ_8 from 0.8405/0.8429 to 0.8271; the cold to baryon ratio stays at 5.411/5.412.

- **The CMB side closes.** The pinned fit is adequate ($-63.79/-63.75$ against a line of 9). PR4 lensing reads $0.9928 \pm 0.0239/0.9929 \pm 0.0239$ of the prediction (-0.30σ on both footings, from $-1.66/-1.89$ unpinned), and DESI DR1 $f\sigma_8$ passes (χ^2 1.95 over four bins).
- **Weak lensing is where it fails.** For form B's spectrum raw KiDS-1000 is best at $\sigma_8 = 0.7358 \pm 0.0162/0.7341 \pm 0.0161$, and the pinned value sits 5.62/5.78 widths above it; with one 0.06 eV neutrino (a bracket at the pinned parameters, not a refit) σ_8 is 0.8152 and the gap is 4.89/5.04 widths. The widths are KiDS's alone. Across every variant below the gap is 5.1 to 5.8 widths, and the lane calls it its robust statement.
- **The label is at its line.** The graded row is an absolute fit. Raw KiDS at the pinned amplitude, on the record's gravity-only run moved by the run's own response, reads χ^2 157.89/159.35 on 104 points, $p = 0.0026/0.0021$, just below the frozen 3σ line of 0.0027. With a constant-factor move (a bracket) the same row reads $p = 0.0044/0.0037$, tension and not exclusion; the reference run alone at the same amplitude reads χ^2 156.88/156.85; with the neutrino bracket $p = 0.0090/0.0077$. Unpinned, the same row reads $p = 0.0004/0.0002$, so the pin moves the form towards the lensing data and not far enough. DES Y3 through the same path reads $p = 0.050/0.051$ (reported, not graded).
- **The exclusion does not survive a check made after grading.** The reference run is a single box of $100\,\mathrm{Mpc}/h$ whose large-scale power is a high draw: its starting spectrum is 1.48 times its own linear spectrum in its third bin. A difference of two models on the same run cancels this; an absolute fit does not. With the realisation taken out in four ways (the run's own starting ratio with two tapers, and two $200\,\mathrm{Mpc}/h$ runs below $k = 0.5\,h\,\mathrm{Mpc}^{-1}$), p is 0.003 to 0.013: tension each time. The check is post hoc, is reported and changes no label; the lane says it would not quote the exclusion without it. CO1's absolute KiDS p-values carry the same noise (its reference row, $p = 0.014$, reads 0.021 to 0.049 corrected); CO1's label rests on a difference and is not affected.
- **What the label covers.** Form B as stated, a gravity-only run with nothing ejecting gas beyond R_{200c}, and the inherited halo list; the fit keeps massless neutrinos and the shear row uses 104 of the 225 published KiDS points. The lane's criteria say, before any number, that a fail here is the weak-lensing amplitude tension that any cold-component fit to the CMB carries: it is not peculiar to form B, and form B does not relieve it. The two lines (0.05 and 3σ) are conventional but were chosen with CO1's numbers in view, which the lane states as a limit. Its one amendment, a known-answer tolerance, is post hoc and was committed before any graded number; the known answer passes the frozen tolerance too.

4 Form B after the regrade

Gate	Reading for form B	Lane
Rotation curves	pass, rms 0.0563/0.0558 (line 0.0660)	PD1
Solar System	pass, carried	PD1
CMB peaks, CMB lensing, growth rate	hold when computed; NO VERDICT with one mutation arm as frozen; redistribution rows on an inherited halo list	F1, CO1
Forest	carried from the cold reference	PD1
Amplitude of fluctuations, pinned	CMB lensing (-0.30σ) and growth pass; raw KiDS prefers σ_8 0.736 against 0.827, 5.1 to 5.8 of its widths; absolute fit excluded at 3σ as frozen (p = 0.0026/0.0021), tension once the reference run's realisation is corrected	CO2
Clusters, amount	not falsified (one-sided); at the ceiling; two-sided gate FAIL	RB1, PD1
Clusters, radial shape	consistent under a post hoc amendment; NO VERDICT as frozen	RB1
Groups	not falsified (one-sided), $f_{\rm ret}$ 0.597; two-sided gate FAIL	RB1, PD1
Bullet	consistent only if the subcluster's stripped gas is counted	RB1
Coma UDGs	consistent on the systematics budget	RB2
Milky Way classical dwarfs	consistent as a class	RB2
Ultra-faints, M31 bright dwarfs	FAIL high	RB2
DF2, DF4, NGC 147, IC 1613	against the form; tides cannot reach them	RB2
KiDS lensing by type	NO VERDICT; early types between the law and the form	KO2
KiDS host-free lenses	passes at the widest veto with neighbours carrying the share; does not separate the form from the law	ONE1
Outer cluster shear	FAIL, about 3σ with systematics	ZN4, FR1
Static state reached by collapse	BREAKS DOWN (version 1.1): the falling clock folds at the centre and the equations give no answer beyond	COL1
Stability of the static state	linearly stable around a point mass and, away from the centre, around isolated discs with gravity coupled; a disc in an external field NO VERDICT with STABLE excluded; the earlier instability belonged to a planar background	SCS1, SCS2, SCS3, TPA1
One dust term for the CMB, clusters and galaxies (version 1.1)	NO VALUE for a quadratic and for an exponential term; keyed on the expansion BREAKS AT TURNAROUND; keyed on the acceleration BREAKS AT A1	MK1, MK2, TK1, AKD1

5 What stays open

1. **The collapse.** For a static clock the action's equations give the law's shape and hold it. No solution takes a free-falling clock, dust falling in with the baryons, to that state. This is the spiral wall of the 2026-11-01 ledger. Version 1.1: lane COL1 computed it and labels it BREAKS DOWN on both footings. A clock that starts as cold dust in the Hubble flow at $z = 10$ around a baryon point mass folds at the centre within 0.063 to 0.154 Gyr, at the three dust constants its brief asked for and at all four masses, and the equations give no answer beyond that moment; one of its known answers fails, so its law-off rows and one bracket are NO VERDICT. Lane FLD1 (DOES NOT on both footings) finds that the folded region today holds every SPARC galaxy's last measured radius (175 of 175) and every galaxy of the high-redshift sample on disk at its stated extent (100 of 100), and stops 1.3 to 5.4 times short of the KiDS windows and 1.1 to 4.2 times short of the form's own cap (canonical). Lane REL1 tried the law on the clock's acceleration relative to the baryons: NOT WELL-DEFINED, since matter's equation of motion becomes fourth order. Lane LFS1 asked whether the clock's leafwise flux can supply the halo: ILL-POSED, and the answer is no; the flux adds no charge.
2. **Stability beyond a point mass.** With gravity coupled the static state is linearly stable around a point mass. Version 1.1: isolated discs are stable too away from their centres (SCS2). A disc in a uniform external field, where the law's phantom density is negative in places, has no verdict (SCS2, SCS3), and SCS3 says that STABLE is excluded. A disc's own centre is open, and the nonlinear outcome has not been computed.
3. **One dust term.** With the quadratic $K(Q)$ the background has $w = w_0/(w_0 + a^3)$ with $w_0 = I_0/4\mu_K^2$. Our arithmetic, not a lane, with H_0 70 and Ω_c 0.26: $w_0 = 1.063 \times 10^{-8}/\mu_K^2$ (μ_K per Mpc), so at $\mu_K = 3.71$ per Mpc the component still has $w = \frac{1}{2}$ at $z = 1089$. Dust by recombination needs μ_K well above that; the

galaxy-lensing bound is 0.1. Version 1.1: lane MK1 labels the question NO VALUE on both footings. The CMB needs μ_K of at least 198 per Mpc for the component to be cold by $z = 3400$ and 2.51×10^4 for its lensing shape, far above anything the galaxy rows allow. MK1 also corrects the bound quoted above: 0.1 is the first failing grid value, and the bound lies between 0.03 and 0.1. Lane MK2 returns NO VALUE for an exponential dust term (as frozen NO VALUE/NO VERDICT). A stiffness keyed on the clock's expansion BREAKS AT TURNAROUND (TK1), and one that falls with the clock's acceleration BREAKS AT A1 (AKD1); Section 7 gives both.

4. **Clusters are not a static-clock state.** Closing X-COP with the static dust term needs μ_K of 1.78/1.74 per Mpc, 17 to 18 times the galaxy bound. No other state has been solved. Version 1.1: MK1 corrects the sentence before: 1.78/1.74 is only where the mean residual is zero, the shape passes at no μ_K, so the dust does not close X-COP at any value, and '17 to 18 times' should read '18 to 60 times'; done as the action's own static equation the mean-zero value is 1.53/1.46. Lane CLS1 searched the action for a cluster state. As frozen its label is STATE EXISTS-PARTIAL, with 0 of 7 clusters holding a state; at the form's own dust constant (198.25 per Mpc and above) it is NO STATE on both footings, proven on all seven.
5. **Groups** are untested from below: form B needs 40 % of their baryons to lie outside R_{500}. Version 1.1: lane GO1 tried the census in group outskirts and is NOT EVALUABLE: the lensing on disk does not reach the turnaround radius.
6. **The richest clusters** sit at or above the form's ceiling, depending on the hydrostatic mass used. Version 1.1: lane XCV1 regraded X-COP with a mass that does not assume hydrostatic equilibrium: NO VERDICT as frozen, since a known answer fails; under its post hoc amendment the class is AT THE CEILING with the central value above it.
7. **The Bullet** passes only with gas credited to the subcluster that no measurement fixes.
8. **Ultra-faints and M31's bright dwarfs** sit above the form. Version 1.1: with the record's systematics budget and the dispersion upper limits counted, neither class's interval excludes zero (DG1); the medians stay high.
9. **DF2 and DF4** sit below it and need a host 17 to 100 times heavier than the form gives. Version 1.1: lane NH2 set NGC 1052's measured mass against that need: NO VERDICT as frozen, since two known answers fail; under its post hoc reading tides cannot reach DF4 and DF2 is not decided.
10. **KiDS.** On all lenses, early types need more beyond the inner window than PD1's truncated halo and less than the law. On host-free lenses the form passes a gate that the law passes too. Discs are not tested by either.
11. **The outer cluster shear** stands near 3σ.
12. **Cosmology.** The weak-lensing amplitude is not solved: the CMB pins σ_8 at 0.8271 and raw KiDS prefers 0.736 for the form's spectrum, 5.1 to 5.8 of its widths lower. Form B states neither gas expelled from haloes nor neutrino mass. The forest row is carried, and the redistribution rows rest on an inherited halo list and on the cluster share being counted inside R_{200c}. Version 1.1: lane KL1 took the gap apart (Section 7). On the surveys' current data the gap is 1.71/1.88 widths on KiDS-Legacy, with an acceptable fit, and 2.18/2.23 on DES Y3 (2.89/2.82 on the frozen scan). These are post hoc readings: the amended label is GAP STAYS and the frozen labels are NO VERDICT.

6 What would decide it

- The spherical collapse of the clock around a baryon mass with the record's own equations was computed in version 1.1 and breaks down (COL1). What would decide the joint now is an equation for what the clock is behind a fold; Section 7 lists what the landed verdicts force on it.
- The stability of the static state for a disc in an external field with converged magnitudes and for an extended baryon distribution, and its nonlinear outcome. Point masses and isolated discs away from their centres are computed and stable.
- The baryons between R_{500} and the turnaround radius around the same groups (thermal SZ, X-ray, stars) with a lensing mass to that radius. Form B needs $f_{\rm ret} = 1$ at turnaround; a census that falls short falsifies it.
- A cluster mass beyond R_{500} that does not assume hydrostatic equilibrium.
- A measured mass for NGC 1052 inside 79 and 165 kpc.
- The dwarf classes regraded with a full systematics budget, frozen before the numbers.
- Lensing by type with more area. A spectroscopic veto with the redshifts on disk has no power: the late-type error at the widest veto is 0.68/0.62 of the law in KO1 and is forecast at 0.57/0.52 with such a veto, against about 0.47 needed.

- For the weak-lensing amplitude, now that the CMB side is pinned: a refit that carries neutrino mass, in place of the bracket at fixed parameters; the shear row on more than one reference run, and on all of the published points; and a statement by the form of whether gas leaves haloes. Version 1.1: KL1 made the refit and put the measured gas in; it finds the single reference run to be the limiting input, so the deciding step is the shear row on reference runs without realisation noise.
- Gaia DR4 (2026-12-02), scored as pre-registered.

7 Version 1.1: the theory in six rules, the rows since, and what a completion must satisfy

The lanes in this section landed on 2026-11-10, after version 1.0, except the earlier lanes named in the survey of media. Each lane froze its criteria first; where a frozen reading and a post hoc reading differ, both are given, the frozen one first. The closure session re-ran DG1's main run (not its mutation twin), all three parts of KL1 and AKD1 from copies, and every label and stored number reproduced apart from run times and memory (`closure_2026_11_10/RERUNS_2026-11-10.md`); a re-run reproduces a lane's arithmetic and is not an independent derivation. KL1's second part was also regraded by lane KL1R, with the same labels. The other lanes named here are quoted from their result files and were not re-run for this note; for the lanes under `fable52_push/`, what their lead session checked is recorded in that folder's `README.md`.

7.1 The theory in six rules

1. **Fields.** The metric, one clock field that picks a slicing, and a polarization on its slices. Matter couples to the metric only. The action is lane QX17's action (A), which is PAPER70's.
2. **One conserved charge.** The clock equation is a conserved current. The universe carries 5.41 units of its charge per unit of baryons, a fitted number. It is not derived: in the action it is an integration constant (note R541, which is not a graded lane), and four attempts to derive it are EXCLUDED on both footings: a clock-matter coupling (R541A), a Gauss-Bonnet source (R541C), a baryon-current coupling (R541D) and a coupling to $B - L$ (R541E). A fifth, an early potential on the clock (R541B), reads TRADES: it moves the number into the potential's height and does not fix it.
3. **Falling clock.** Where the clock free-falls with the expansion the charge is cold dust. This is the cold component of cosmology.
4. **Standing clock.** Where the clock stands still around baryons its own equations give the law, with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. The halo is the charge itself and gravitates and lenses as mass.
5. **Counted once.** A halo is made of the charge its region brought. A hot system holds 5.41 times the baryons that fell into it. A body that brought no charge has no halo.
6. **The joint.** How a falling clock becomes a standing one is not derived.

The data rows of this note are computed from rules 3 to 5, and several of them fail or have no verdict. Rule 6 is the missing equation, and the hot branch of rule 5 is a rule and not a solution of the action.

7.2 What the landed verdicts force on the joint

Any completion has to meet all of these. Each line rests on a lane's verdict on both footings; where a sentence is a reported reading and not a label, it says so.

1. **Not a wall at a fixed place.** HE1 FAILS, and PAPER71's switch class is closed.
2. **Not a dust term that depends on the clock's offset alone.** MK1 and MK2 return NO VALUE (Section 5, item 3). The offset is about 10^{-6} both in the background at $z = 3400$ and in a galaxy's well today.
3. **Not the expansion alone.** Lane TK1 keyed the dust stiffness on the clock's expansion: BREAKS AT TURNAROUND. The key keeps the expanding background well posed, with a margin of 1.19 on the Jeans line. A collapsing region passes through small expansion while its clock is still falling; there short waves grow without bound, 2 to 140 Myr before the turnaround, and the region never turns around.
4. **Not a stiffness that falls with the clock's acceleration.** Lane AKD1: BREAKS AT A1. The key is a negative copy of the law's own term. Its weight against the law is $3.9 \times 10^{14}/4.2 \times 10^{14}$ at $z = 3400$ and $4.5 \times 10^{11}/4.9 \times 10^{11}$ at $z = 1100$, so the clock's inertia is negative on every sub-horizon scale there, until

$z = 11.6/11.4$. A shape that is flat at zero acceleration has the same sign problem in the outskirts of static wells; that is a reported class bound and not a label.

5. **Not the law on the clock's acceleration relative to the baryons.** REL1: NOT WELL-DEFINED.
6. **It has to survive the fold.** COL1: BREAKS DOWN. Nothing in the action removes the infall energy.
7. **It has to reach further than today's fold.** FLD1: DOES NOT.
8. **It cannot draw the halo from the clock's leafwise flux, and it needs a cluster state the action does not have.** LFS1: ILL-POSED. CLS1: NO STATE at the form's own dust constant.

So no dust term tried that is built from the clock's own first-order kinematics, its offset, its expansion or its acceleration, joins the two states. Of functions of the offset, only a quadratic and an exponential were graded; MK1's statement about any function of the offset alone is a requirement and not a graded result. One kinetic function with a narrow interval of changed compressibility is named in the two assembly notes: its linear and static checks pass there, and whether a forming halo can reach its static state is not computed. It is not a result.

7.3 Media with more structure than one clock

A design note, `closure_2026_11_10/MEDIUM_DESIGN_2026-11-10.md`, which is not a lane, sets the classes of medium the record has tried against four properties the data ask for: conserved and cold; passing through itself; hot and at full share in clusters; locked to the baryons in galaxies. Each class has a failing verdict in its own lane: any local, momentum-conserving, normally self-gravitating medium whose equilibrium is the law (M5: NO-GO, a theorem); two phases (PH1: NO-GO); a superfluid with phonons (QX28: FAILS at G3, G4, G5, G6 and G8); a self-interacting isothermal inner halo (IJ1: FAILS); a light fermion beside the law (HF1: fails its first gate at 5 of 6 masses with 1 undetermined; the lane says this kills the family on its inputs and is not a no-go theorem) and a warm component beside it (WM1: FAIL AT SEEDING, a gate added after its first result, which failed on groups); the tethered medium (TM1: it does not resolve law against clumping, and its scored rows fail or are conditional); and nonlocal, history and global actions (FA3: no candidate survives). Route E, in which cold dark energy clumps everywhere, gravity is Newtonian and the law is emergent, had its cold sector retired by the owner on 2026-11-01.

One class had no lane: a medium in thermal contact with the baryons where their acceleration exceeds a_0. The note works it out as our own arithmetic, not a graded result. The contact needs an opacity $\pi G/a_0$ of $22.4/18.5\,\mathrm{cm^2\,g^{-1}}$. With it the medium's mean free path against baryons at $z = 1100$ is 25.8/31.1 pc and its drag at a relative speed of $30\,\mathrm{km\,s^{-1}}$ is 0.73/0.61 of the expansion rate, so it does not stay decoupled from the plasma; and a cluster's gas is 0.14 to 0.68 optical depths (canonical), so it would be dragged with the Bullet's gas.

7.4 The data rows since version 1.0

- **Weak-lensing amplitude (KL1).** CO2's gap of 5.62/5.78 KiDS-1000 widths came apart into pieces. The largest is the survey's own release: for the form's spectrum KiDS-Legacy prefers $\sigma_8 = 0.7767/0.7748$ where KiDS-1000 preferred 0.7358/0.7341. The largest on our side is the massless neutrinos: a refit of the CMB fit with one 0.06 eV neutrino gives $\sigma_8 = 0.8121/0.8120$. The covariance evaluated at the model and the gas RB1 measured outside R_{500} are smaller pieces. As frozen, the lane's first two parts read NO VERDICT, because a mutation arm cannot bite on DES Y3. Under a post hoc amendment, KiDS-1000 with DES Y3 reads TENSION (KiDS-1000 gap 3.95/4.06 widths) and KiDS-Legacy with DES Y3 reads AMPLITUDE CONSISTENT WITH RAW LENSING ($p = 0.111/0.101$ for Legacy), with Legacy still 2.13/2.30 of its widths below the form, which the lane's frozen list counts against the form. Lane KL1R, a regrade of that part by another session, returns the same two readings; it says its blindness was not clean, since KL1's headline numbers were visible before it began, and that it is not an independent check of the shared machinery.
- **The second pass.** With the reference run's known draw corrected and the cold component responding to the moved gas, the lane reads NO VERDICT as frozen (a mutation arm does not bite, and on alt a known answer fails on DES Y3) and, under three post hoc amendments, GAP STAYS, driven by DES Y3: KiDS-Legacy 1.71/1.88 widths and consistent (2.06/2.23 with the full Gaussian likelihood), DES Y3 2.18/2.23 on the polished scan (2.89/2.82 on the frozen scan), KiDS-1000 3.72/3.85. Held to Legacy's own alignment model under its own prior, Legacy reads 1.57/1.83 and stays consistent. Gas taken from a stellar-mass-selected stack is a bracket and barely moves anything. The lane concludes that the remaining gap is the size of the reference run's realisation noise: two independent $200\,\mathrm{Mpc}/h$ runs spliced in at low k put DES Y3 at 1.31/1.27 against 3.49/3.43 widths. [PENDING: BX1: the gate re-graded with fixed-amplitude reference runs; label and origin

commit]

- **Hosted dwarfs (DG1): HISTORY DOES NOT CLOSE THEM on both footings.** RB2's two failing classes rested on measurement scatter alone. With the record's joint systematics budget re-derived for the form and the dispersion upper limits counted, both intervals contain zero on both footings: the ultra-faints read +0.550 [−0.090, +1.175] (alt +0.509 [−0.110, +1.141]) and M31's bright dwarfs +0.299 [−0.181, +0.774] (alt +0.258 [−0.237, +0.747]). The medians stay high; this is the standing RB2 gave Coma's ultra-diffuse galaxies and not a pass. A gas-history reading, in which a dwarf keeps the halo of the gas it lost, brings the class medians to −0.045, +0.037 and −0.010 dex and fails the per-galaxy gates: inside the ultra-faints the rank correlation of the residual with the predicted history is +0.06, and a two-number stellar-mass power law fits better (rms 0.306 against 0.343 dex). The lane is post hoc to RB2's class numbers and says so.
- **The cold component's shape (SHP1).** Its label is THE LAW'S SHAPE ON ROTATION CURVES, LENSING ABOVE; NOT ONE SHAPE (R3b) on both footings. On rotation curves (2700 points in 149 galaxies) everything that is not baryons follows the law's shape in every bin at the standard mass-to-light ratios, the worst bin median within 0.057 dex; at disc ratios of 0.3 or 0.8 that pass fails. KiDS isolated lenses have the same shape at $1.40 \pm 0.03/1.28 \pm 0.03$ times the law with the published stars only, and sit below it with the published hot gas counted as baryons. A separate halo of PD1's 59.5 M_b on top of the law is excluded on every reading.

8 Our own estimates

Recorded before the numbers that tested them. (i) The KO2 brief put the error on the disc ratio near 0.2 and expected discs low and early types at 0.7 to 1.0. A model-only toy at the freeze gave the brief's ratio about twice the error of its replacement (0.37 against 0.19), so the lane replaced the primary statistic before computing; the brief's ratio then measured an error of 0.278 on discs, and discs came out undecided. (ii) The expectation that discs carry no outer field rested on the host-free lead, which does not generalise to the full sample. (iii) The RB1 brief expected X-COP near 0.96; it reads 1.095 on the forward-fitted mass. (iv) RB2 expected M31's bright dwarfs to be borderline; they fail. (v) The Bullet estimate, closer than 2.6σ once the stripped gas is counted, held; so did the group range and the Coma shift. (vi) CO2's hand estimate, written into its criteria before any code, put the optical depth at 0.055 to 0.060, σ_8 near 0.828, PR4 lensing within 1σ and raw KiDS near the line between tension and exclusion. The optical depth came out just below that range (0.0532); the rest held. (vii) The lead session and TPA1 expected the instability to survive the coupling to gravity, and the draft of this note carried it as the static state's own. It did not survive: SCS1's expectation before its freeze, stable with the earlier result planar, is the outcome. Version 1.1: (viii) The closure session expected the survey's newer release to cut the weak-lensing gap from about five widths to one or two, and the neutrino floor, the covariance and the gas together to be worth about one more. The release cut the σ_8 gap by about 45 % and left 4.4/4.6 of Legacy's narrower widths; the three inputs were worth 2.3 more widths on Legacy and 1.7 on KiDS-1000. (ix) It expected gas from the stellar-mass-selected stack to be worth 0.5 to 1 widths; it was worth 0.05 to 0.09 and only a bracket. (x) It proposed that the dwarfs that sit high are the ones that lost their gas. DG1 found the class medians matched and the per-galaxy gates failed. (xi) It proposed the acceleration key and wrote that its gradient term had a healthy sign. AKD1 found the sign the other way: the term is kinetic, with a negative coefficient.

Method

Lanes KO2, RB1, RB2, CO1 and CO2 are under `closure_2026_11_09/`; HE1, ONE1, TPA1, SCS1 and the notes FB1 and KO3F are under `fable52_push/`. None is registered in the closure record and none relabels an earlier lane. Post hoc items are marked in their lanes: RB1's amendment on the cluster shape gate, KO2's four amendments (the third, form B's prediction, was committed before any fitted amplitude was opened), KO2's mutation twin that did not bite, CO1's restated mutation arm, CO2's tolerance amendment, CO2's realisation check, which was made after grading, and SCS1's three amendments. RB2 has no amendment. This note claims no Lean theorem of its own; the Lean build that earlier lanes compiled with was removed from the machine on the evening of 2026-11-09 and a build inside the repository has been used since 22:17 that evening, so a Lean file's status is what its lane's compile output says. Distances for the KiDS lenses use the record's fiducial flat background (Ω_m 0.3, H_0 70), as in the staged sums. No halo model, no concentration relation and no simulation fit enters a graded row of KO2, RB1 or RB2; one externally calibrated bracket in RB1 is labelled as such. CO1's redistribution rows,

which CO2's weak-lensing row reuses, inherit a mass function, a bias and a concentration relation calibrated on simulations, and say so. CO2's low-multipole table is Planck 2018 and its high-multipole spectra are PR4. The owner's decision is recorded in `closure_2026_11_09/OWNER_DECISION_2026-11-09_FINAL_FORM_B.md`. The supplement holds every lane file.

Version 1.1's lanes: SCS2, SCS3, DCN1, MK1, MK2, COL1, FLD1, REL1 and SHP1 are under `fable52_push/`; TK1, AKD1, LFS1, CLS1, KL1, KL1R, DG1, NH2, XCV1, GO1, the note R541 and its lanes R541A to R541E are under `closure_2026_11_10/`, beside the notes `THEORY_ASSEMBLED_2026-11-10.md` and `MEDIUM_DESIGN_2026-11-10.md`, which are assemblies and grade nothing. None is registered and none relabels an earlier lane. Post hoc items are marked in their lanes: KL1's amendments A1, C1, C2 and C3, on which its readings other than NO VERDICT rest; DG1's amendment; TK1's, FLD1's and REL1's amendments; COL1's four amendments; MK2's second, third and fourth amendments; SCS2's second to fourth, SCS3's four and DCN1's six; GO1's amendment; SHP1's check made after grading; and the readings of NH2 and XCV1 other than NO VERDICT. WM1's seeding gate was added after its first result. AKD1 has no amendment; its extra readings were added after the freeze and set no label. The earlier lanes named in Section 7's survey of media (M5, PH1, IJ1, HF1 and TM1 under `rebuild_2026_10_22/`, QX28, FA3 and WM1 under the closure folders of 2026-11-06, 11-07 and 11-09) are quoted by the first lines of their result files only; those files are added to the supplement. The KiDS-Legacy release was fetched on the owner's go and sits beside the repository, not in it.